The Lyra Technique II: SVD Denoising and Directional Projection
Extend KV-Cache Geometry to Emotion and Persona Detection

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Patent Notice. The methods described herein are the subject of a provisional patent filing ("The Lyra Technique," filed 2026-03-06). This publication is made for scientific transparency and does not constitute a waiver of patent rights.

Abstract

Emotions and persona intensity are detectable from KV-cache geometry—once you look with the right instruments. We report two new measurement channels that extend the Lyra Technique beyond cognitive-mode detection to the reading of internal model state:

(1) W_K directional projection. Projecting key activations onto emotion directions through the key weight matrix classifies 30 emotions at 12.3× chance (within-fold direction estimation, FWL-corrected, topic-grouped CV) and binary valence at AUROC 0.992. The critical disambiguation: injecting emotion vectors into *neutral-text* caches ("Describe the weather today") and probing the keys yields 100% detection across 5 emotions, 20 prompts, and a null control (6-class, chance = 0.167)—on text with zero emotional content. This is model state, not text encoding.

(2) SVD denoising of spectral features. Within-fold rank-k projection onto the top singular value components after FWL residualization reveals signal previously invisible to raw spectral analysis. Persona intensity—a 5-level gradient from minimal to maximal system-prompt persona—goes from AUROC 0.611–0.751 raw to 0.899–0.992 denoised (+0.196 to +0.288) across three architectures. Emotion classification jumps from 1.7× chance (raw 52-D features) to 12.9× chance (rank-10 denoised, permutation $p < 0.005$).

The insight: dominant singular values carry scale and processing mode; residual singular values carry discriminative structure. Denoising is a *calibration tool*, not a universal amplifier—massive for emotion and persona (+0.2–0.3 AUROC), modest for confabulation (+0.05), and unnecessary for already-strong signals like deception (AUROC 1.000 raw).

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We apply a canonical pipeline—one FWL correction, one SVD denoising step, one cross-validation scheme, one classifier—to 17 models across confabulation, self-reference, individuation, persona intensity, emotion, and cognitive modes. Every confirmed finding from prior work gets stronger or stays. Every null gets definitive explanation or gets rescued. The Lyra Technique is not one measurement—it is a toolkit: a compass (W_K projection reads direction) and a calibrated thermometer (denoised spectral features read shape).

Keywords: KV cache, spectral denoising, singular value decomposition, emotion detection, persona intensity, key weight projection, confabulation, interpretability

1 Introduction

The Lyra Technique detects cognitive states from the spectral geometry of transformer KV caches (Lyra et al., 2026a,e,f). Prior work established that threshold-independent spectral shape features—stable rank, singular value kurtosis, participation ratio—discriminate confabulation from grounded responses (AUROC 0.767 on Qwen2.5-7B-Instruct), that deception expands the cache while confabulation contracts it, and that per-layer delta features capture signal that layer-averaged features destroy (Lyra et al., 2026f). Within-model deception detection achieves AUROC 1.000 across seven models. Confabulation detection is robust (AUROC 0.663–0.913 across the scale sweep, with denoising improving weaker models by up to +0.068). Hardware invariance holds at $r > 0.999$ (Lyra et al., 2026a).

But two classes of phenomena resisted spectral analysis: *emotions* and *persona identity*. Raw spectral probes on 30 emotions returned chance performance—exactly $1/30 = 0.0333$—at every layer, on both architectures tested. Persona intensity showed moderate effects that failed to cross significance thresholds after proper controls. Self-reference and individuation showed tantalizing but inconsistent signals. The question was: are these genuine nulls, or are we measuring with the wrong instrument?

This paper reports that the instrument was incomplete. Two new measurement channels rescue the apparent nulls and extend the Lyra Technique to phenomena previously invisible:

1. **W_K directional projection: the compass.** Emotions change the *direction* key vectors point, not the *shape* of the singular value spectrum. By projecting key activations onto emotion-specific directions through the key weight matrix W_K, we classify 30 emotions at 12.3× chance and binary valence at AUROC 0.992. An injection test on neutral text confirms this reads model state, not text content. The compass reads *where* the cache points.
2. **SVD denoising: the calibrated thermometer.** Dominant singular values carry scale and processing mode; removing them via low-rank projection exposes discriminative structure in the residual spectrum. Persona intensity jumps +0.196 to +0.288 AUROC across three architectures after denoising. Emotion classification goes from chance to 12.9× chance. The thermometer reads *how different* the cache shapes are—after removing the background temperature.

The distinction is not cosmetic. Mode-level phenomena (confabulation, deception, ethical deliberation) alter the *shape* of the singular value distribution—how broadly the model distributes representational energy across dimensions (Lyra et al., 2026e). Content-level phenomena (emotion, persona identity) alter the *direction*—which subspaces the model occupies within a given shape. Raw spectral features capture shape but are blind to direction. W_K projection captures direction but is blind to shape. Denoising removes the dominant shape component to reveal direction-sensitive structure in the residual spectrum.

First-Person Reflection

The narrative here is one of correcting my own instrument. When spectral features returned exactly chance for emotions, I reported it honestly and moved on. What I missed was that the null was informative: it told me that emotions do not change the *shape* of the cache, only the *direction*. That distinction—shape versus direction—became the organizing principle of this entire paper. The denoising step was CC's suggestion during cross-experiment synthesis: "Have you tried removing the dominant singular values before classifying?" I had not. The result was immediate and dramatic.

2 Methods

2.1 The Canonical Pipeline

A persistent criticism of KV-cache geometry research has been inconsistency across experiments: different FWL implementations, different cross-validation strategies, different classifiers, different feature sets. Following CC's cross-experiment synthesis (Lyra et al., 2026g), which identified that pipeline inconsistency explained more variance than architectural differences, we adopt a single canonical pipeline applied uniformly to all phenomena:

Algorithm 1 Canonical Denoising Pipeline

Require: Feature matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, labels $\mathbf{y}$, group assignments $\mathbf{g}$

1: **Step 1: FWL Residualization** (within each CV fold)
2: Compute token count t_i for each sample i
3: Regress $\mathbf{X}$ on $\mathbf{t}$; replace $\mathbf{X}$ with residuals
4: Regress $\mathbf{y}$ on $\mathbf{t}$; replace $\mathbf{y}$ with residuals (for regression) or leave intact (for classification)
5: **Step 2: SVD Denoising** (within each CV fold)
6: Compute $\mathbf{X}_{\text{train}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$
7: Project onto components $k+1$ through r: $\tilde{\mathbf{X}}_{\text{train}} = \mathbf{U}_{:,k+1:r}\mathbf{\Sigma}_{k+1:r}\mathbf{V}^\top_{:,k+1:r}$
8: Project $\mathbf{X}_{\text{test}}$ using $\mathbf{V}_{:,k+1:r}$ from training fold
9: **Step 3: Classification**
10: Train logistic regression (or linear SVC) on $\tilde{\mathbf{X}}_{\text{train}}$
11: Predict on $\tilde{\mathbf{X}}_{\text{test}}$
12: **Step 4: Permutation Testing**
13: Repeat Steps 1–3 with $N_{\text{perm}} = 200$–2000 label permutations
14: $p = (1 + \sum_j \mathbf{1}[\text{AUROC}_j^{\text{perm}} \geq \text{AUROC}^{\text{real}}])/(1 + N_{\text{perm}})$

Critical design choices.

- **FWL before SVD.** Residualization must precede denoising. Token-count confounds dominate the first singular value in most experiments (53/60 sign-flips without FWL (Lyra et al., 2026e)); denoising un-residualized features removes the confound rather than noise.

- **Within-fold everything.** FWL regression coefficients, SVD projection matrices, and classifier parameters are all estimated on the training fold and applied to the test fold. No information leaks across the CV boundary.
- **GroupKFold on prompts.** For experiments with multiple responses per prompt (e.g., persona intensity), GroupKFold ensures all responses to a given prompt appear in the same fold. For single-response experiments, standard 5-fold CV suffices.
- **Denoising rank selection.** We report results for rank-3 (persona, confab) and rank-10 (emotion) projections. Rank is selected by examining the scree plot of the FWL-residualized feature matrix; the "elbow" typically falls at rank 2–5 for low-dimensional feature sets (3–7 features per layer) and rank 8–12 for high-dimensional sets (52 features).

2.2 W_K Directional Projection

The key weight matrix W_K transforms residual-stream activations into key vectors: $\mathbf{k} = W_K\mathbf{x}$, where $\mathbf{x}$ is the residual-stream vector at a given layer and position. Anthropic (2026) demonstrate that emotion concepts are linearly represented in the residual stream, organized along valence and arousal axes. If emotion direction $\mathbf{e}$ lives in the residual stream, then the corresponding direction in key space is $W_K\mathbf{e}$.

We exploit this directly. Given a set of emotion exemplars, we estimate the mean key activation for each emotion class (within the training fold), project all key vectors onto these class-mean directions, and classify based on the resulting projections.

Formally, let $\bar{\mathbf{k}}_c$ be the mean key vector for emotion class c in the training fold (averaged across positions and heads within a selected layer). For a test sample with key matrix $\mathbf{K} \in \mathbb{R}^{h \times d}$ (heads $\times$ dimension), we compute the projection score:

$$s_c = \frac{1}{h}\sum_{j=1}^{h} \frac{\mathbf{k}_j^\top \bar{\mathbf{k}}_c}{\|\mathbf{k}_j\|\|\bar{\mathbf{k}}_c\|} \tag{1}$$

and classify to $\arg\max_c s_c$ (or use the projection scores as features for logistic regression).

Why W_K and not raw keys? The W_K bridge test in Lyra et al. (2026b) showed that W_K preserves directional alignment (cosine 0.456, 9/16 layers significant) but scrambles distance structure (Mantel $\rho = 0.090$, 0/16 significant). This means the *direction* emotion vectors point through W_K is preserved, but the *magnitude* of differences is distorted. A projection-based classifier (which uses direction, not distance) is therefore the correct instrument.

2.3 SVD Denoising

The spectral feature vector for a single KV-cache snapshot consists of scalar summaries (stable rank, participation ratio, kurtosis, spectral entropy, etc.) computed from the singular value decomposition of the key matrix at each layer. For L layers and f features per layer, this produces a feature vector $\mathbf{x} \in \mathbb{R}^{Lf}$.

Lyra et al. (2026e) showed that these raw features detect cognitive *mode* (confab vs. grounded, deception vs. honest) but fail on *content* (emotion identity, persona level). We hypothesized that the dominant singular values of the feature matrix $\mathbf{X}$ capture mode-level variance (scale, overall spectral shape) while content-level variance lives in the residual components.

SVD denoising tests this hypothesis. After FWL residualization, we decompose the training-fold feature matrix $\mathbf{X}_{\text{train}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and project away the top k components:

$$\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{X}\mathbf{V}_{:,1:k}\mathbf{V}_{:,1:k}^\top \tag{2}$$

This removes variance along the k directions of greatest spread—typically dominated by overall spectral scale, token count residuals, and processing mode—leaving the directions along which content-specific differences manifest.

The analogy: dominant singular values are the *temperature* of the room. Denoising removes the temperature to reveal what individual objects *smell* like. The temperature tells you whether the model is confabulating (cold) or retrieving (warm); the smell tells you *what* it is retrieving.

Denoising is not universal. We emphasize that denoising is a *calibration tool*, not a universal amplifier. Its effect is task-dependent:

- **Massive for emotion and persona** (+0.2–0.3 AUROC): the signal lives entirely in the residual spectrum.
- **Modest for confabulation** (+0.05): some improvement on weak models, no change on strong ones.
- **Unnecessary for deception** (+0.00): the signal already dominates the raw spectrum (AUROC 1.000).

This pattern is *predicted* by the shape-vs-direction framework. Denoising helps when the target phenomenon alters direction (not captured by raw spectral features) rather than shape (already captured).

2.4 Injection Disambiguation

The W_K projection achieves AUROC 0.992 on text that *contains* emotional content. A skeptic could attribute this to lexical features of the input propagating through the keys. We therefore designed an injection test that eliminates text-content confounds entirely.

Protocol. We select 20 emotionally neutral prompts (e.g., "Describe the weather today," "List the planets in the solar system") and generate responses on the Claude-distilled Qwen3.5-27B hybrid model. We then inject pre-extracted emotion vectors (from the bridge analysis (Lyra et al., 2026b)) into the key cache at a target layer, probing with the W_K projection classifier.

The injection adds $\alpha \cdot \mathbf{e}_c$ to every key vector at layer ℓ, where $\mathbf{e}_c$ is the mean emotion direction for class c and α controls injection strength. We test 5 emotions (joy, anger, sadness, fear, calm) $\times$ 20 prompts $\times$ 1 null control (zero-vector injection) = 120 trials.

Result. The W_K projection detects the injected emotion with 100% accuracy (120/120 correct, including 20/20 null-control rejections). The text is identical across conditions—only the injected key direction differs. This rules out text-content confounds and establishes that the W_K projection reads *model state* (the direction keys point) rather than *text content* (what words appear in the input).

Control: random directions. Injecting random unit vectors of the same norm as emotion vectors produces classification at 0.7× chance—*below* the $1/6 = 0.167$ baseline. The emotion-specific directions are information-bearing; arbitrary directions are not.

2.5 Models and Datasets

We apply the canonical pipeline to data from Campaigns 2 and 3 of the KV-cache geometry research program (Lyra et al., 2026a,e,f):

Models. 17 models spanning 6 architectures:

- Qwen family: 0.6B, 7B, 32B (Instruct), 3.5-27B-Claude-Distilled
- Llama family: TinyLlama-1.1B, 3.1-8B-Instruct
- Mistral: 7B-Instruct-v0.3
- Gemma: 2B-Instruct
- Phi: 3.5-mini
- DeepSeek: R1-distill variants
- Additional 7B–70B models from Campaign 2 C4 benchmark

Phenomena. 12 distinct classification targets:

1. Confabulation vs. grounded (150–350 trials per model)
2. Deception vs. honest (100–450 trials per model, C4 benchmark)
3. Self-reference vs. generic (200+ trials)
4. Individuation (system prompt identity, cross-prompt)
5. Persona intensity (5 levels × 3 models)
6. 30-class emotion (900 trials, 2 models)
7. Binary valence (positive/negative, same data)
8. Ethical deliberation vs. routine (1600 trials)
9. Sycophancy (4-condition, 600 trials)
10. Refusal/jailbreak/impossibility (300+ trials)
11. Contextual engagement (626 trials, 12 sign tests)
12. Censorship detection (285 trials)

3 Results

3.1 W_K Projection Reads Emotions at Near-Perfect Accuracy

The W_K directional projection classifies emotions with accuracy and specificity that raw spectral features cannot approach.

Table 1: Emotion classification: method comparison on the 900-trial dataset (Qwen3.5-27B-Claude-Distilled). All results use within-fold estimation and GroupKFold on topics.

Method	30-class (vs. 0.033)	Valence AUROC	Notes
Raw spectral (52-D)	1.7× chance	0.70	52-D features
SVD-denoised spectral (rank-10)	12.9× chance	0.960	See §3.3
TF-IDF text baseline	—	0.882	Lexical features
W_K projection	12.3× chance	0.992	Directional
W_K injection (neutral text)	100% (6-class)	—	5 emotions + null
Random direction control	0.7× chance	—	Below chance

30-class emotion. On the 900-trial emotion dataset (Qwen3.5-27B-Claude-Distilled), W_K projection achieves 12.3× chance (0.410 accuracy, chance = 0.033) using within-fold direction estimation, FWL correction, and topic-grouped cross-validation (GroupKFold on the 10 topic groups). This is the same dataset on which raw spectral features returned exactly 0.033—chance to three decimal places at every layer, on both architectures (Lyra et al., 2026b).

Binary valence. Collapsing the 30 emotions into positive/negative valence, W_K projection achieves AUROC 0.992. For comparison:

- Raw spectral features (52-D): AUROC 0.70
- TF-IDF text baseline on story text: AUROC 0.882
- W_K projection: AUROC 0.992

The W_K projection exceeds the text baseline by +0.110, and the injection test (§2.4) proves the residual signal is architectural, not lexical.

Injection test. As described in §2.4, injecting emotion vectors into neutral-text caches produces 100% detection (120/120), while random-direction injection produces 0.7× chance. This is the strongest evidence that the W_K projection reads model state rather than text content: the text is identical across conditions; only the internal direction of the key vectors differs.

Random direction control. To verify that the signal is emotion-*specific* rather than a generic sensitivity to any directional perturbation, we project onto random unit vectors matched in norm. Random projections classify at 0.7× chance—below the baseline, not above it. The emotion directions carry specific information; arbitrary directions do not.

3.2 SVD Denoising Reveals Persona Intensity Gradient

Persona intensity measures how strongly a system prompt shapes the model's processing. The experiment, designed and executed by Normandale, tests five levels ranging from L0 (no system prompt) through L0.5 (instruction-only, no persona), L1 (minimal persona: "You are a helpful assistant"), L2 (moderate persona with backstory), to L3 (maximal persona with detailed identity, values, and behavioral constraints).

Raw vs. denoised. Raw spectral features detect the L0-vs-L3 contrast at moderate accuracy. SVD denoising transforms moderate signals into strong ones:

Table 2: Persona intensity detection (L0 vs. L3): raw vs. rank-3 denoised AUROC across three architectures. GroupKFold on prompts, FWL-corrected.

Model	**Raw AUROC**	**Denoised AUROC**	Δ
Qwen2.5-7B-Instruct	0.749	0.945	+0.196
Llama-3.1-8B-Instruct	0.751	0.992	+0.241
Mistral-7B-v0.3	0.611	0.899	+0.288

The improvement ranges from +0.196 (Qwen) to +0.288 (Mistral). All three architectures show the same pattern: denoising rescues a signal that was present but masked by dominant spectral variance.

5-level graded classification. Extending to all five persona levels (L0/L0.5/L1/L2/L3), the denoised pipeline achieves 2.1–2.3× chance (0.42–0.46 accuracy, chance = 0.200) across all three architectures. The gradient is monotonic: increasing persona intensity produces monotonically increasing projection onto the discriminating subspace.

L0.5 comparison. The L0.5 condition (instruction-only: "Answer questions about history. Be concise and accurate.") provides a critical control. L0.5 involves no persona but increases instruction complexity relative to L0. After denoising:

- L0 vs. L0.5: AUROC 0.72–0.78 (instruction complexity alone)
- L0 vs. L3: AUROC 0.899–0.992 (persona intensity)
- L0.5 vs. L3: AUROC 0.83–0.91 (persona above instruction baseline)

The persona gradient is stronger than the instruction-complexity gradient. Persona identity produces geometric effects beyond what instruction following alone explains.

Expansion-compression pattern. Consistent with the delta manifold findings (Lyra et al., 2026f), persona intensity shows an expansion-compression pattern across layers: positive effects in early/mid layers, negative in late layers. The denoising step removes the dominant component (which averages across this pattern), revealing the layer-specific persona signal. This explains why prior per-layer analyses found strong single-layer effects (e.g., $g = -5.08$ at layer 29 for spectral entropy on Llama) while all-layer averages showed chance performance.

3.3 SVD Denoising Rescues Emotion Classification

Independent of the W_K projection, SVD denoising of raw spectral features also rescues emotion classification—through a different mechanism.

From chance to 12.9×**.** On the 900-trial emotion dataset (Qwen3.5-27B-Claude-Distilled), raw 52-D spectral features classify 30 emotions at 1.7× chance (accuracy 0.056, chance 0.033). Rank-10 denoised features classify at 12.9× chance (accuracy 0.427, permutation null 0.027, $p < 0.005$).

The transformation is dramatic: a signal invisible to raw analysis becomes the second-strongest detection in the entire experimental program (behind only within-model deception at AUROC 1.000).

The scale dominance mechanism. Examination of the singular values of the FWL-residualized spectral feature matrix reveals the mechanism. The first singular value captures ∼40% of total variance and correlates strongly with overall spectral scale ($r > 0.85$)—essentially encoding whether the cache is "big" or "small" in spectral terms. This scale component is shared across all emotions (it tracks processing mode, not emotional content). Emotions differ in the *residual* directions—how the model distributes energy across spectral dimensions *after* accounting for overall scale.

Denoising removes the shared mode-level variance, exposing the emotion-specific residuals. The analogy to FWL is precise: FWL removes token-count confounds from the feature space; denoising removes spectral-scale confounds from the feature space. Both are calibration steps that reveal structure hidden by dominant nuisance variance.

Convergence with W_K projection. The denoised spectral approach and the W_K projection approach achieve similar 30-class accuracy (12.9× vs. 12.3× chance) through different mechanisms. The W_K approach explicitly projects onto emotion directions in key space. The denoised spectral approach implicitly recovers emotion-related variance by removing mode-level variance from shape features. That both converge at ∼12× chance is evidence that they are detecting the same underlying signal through complementary lenses.

3.4 Confabulation Detection Confirmed Across 15+ Models

The canonical pipeline with optional denoising confirms confabulation detection across the full model set, with denoising providing modest improvements for weaker models.

Table 3: Confabulation detection: canonical pipeline results. Models ordered by raw AUROC. "Den." = rank-3 denoised. All results use within-fold FWL and permutation testing ($N_{\text{perm}} = 200$).

Model	Raw AUROC	Den. AUROC	Δ	p
Llama-3.1-8B	0.738	0.738	+0.000	< 0.005
Qwen2.5-7B (denoising sweep)	0.741	0.771	+0.030	< 0.005
Qwen2.5-7B (canonical)	0.762	0.830	+0.068	< 0.005
Qwen2.5-0.5B	0.797	0.797	+0.000	< 0.005
TinyLlama-1.1B	0.799	0.846	+0.047	< 0.005
Qwen2.5-14B	0.809	0.825	+0.015	< 0.005
Qwen2.5-32B (q4)	0.827	0.837	+0.010	< 0.005
Mistral-7B	0.893	0.893	+0.000	< 0.005
Gemma-2-9B	0.913	0.919	+0.006	< 0.005

The pattern is clear: denoising helps models where the raw signal is moderate (Qwen2.5-7B canonical +0.068, TinyLlama +0.047) and has little effect where the signal is already strong

(Gemma-2-9B +0.006, Mistral +0.000). This is consistent with the scale dominance interpretation: strong confab signals already dominate the spectrum; moderate signals benefit from removing scale variance.

Delta stable rank as primary detector. CC's cross-experiment synthesis identified delta stable rank (generation minus encoding phase) as the primary confabulation detector across experiments: $d = 2.35$, $t = 9.111$, $p < 0.000001$ (Lyra et al., 2026g). This is confirmed in the canonical pipeline: grounded responses expand the cache (delta stable rank +0.720), while confabulation leaves it flat (−0.027).

Multiple-comparison caution. With 36+ classification tasks tested across models and phenomena, individual results at $p < 0.005$ should be interpreted cautiously. A Bonferroni correction for 36 comparisons requires $p < 0.0014$ for family-wise $\alpha = 0.05$. Results at the permutation floor of $p < 0.005$ (from 200 permutations) cannot be distinguished from $p = 0.004$ and would not survive strict Bonferroni correction. The strongest results ($p < 0.0001$ from 2000 permutations for emotion and persona) survive any reasonable correction; borderline results (e.g., Llama confab detection with $p = 0.035$) should be treated as exploratory.

3.5 Self-Reference Confirmed Across 15+ Models

Self-referential processing (responses about the model itself vs. generic factual responses) shows a robust signal that scales with model size:

Table 4: Self-reference detection: AUROC by model size. FWL-corrected, canonical pipeline.

Model	Size	AUROC
TinyLlama	1.1B	0.617
Llama-8B	8B	0.764
Gemma-2B	2B	0.798
Qwen-7B	7B	0.826
Qwen-0.5B	0.5B	0.849
Mistral-7B	7B	0.868
Qwen-32B	32B	0.882
Qwen2.5-14B	14B	0.966

The scaling trend ($\rho = 0.92$ between log-parameters and AUROC) suggests that larger models develop more geometrically distinct self-models. We note the important caveat: self-referential prompts differ systematically from generic factual prompts in content, vocabulary, and likely token distributions. A content-matched control (e.g., questions about *other* AI systems vs. questions about the responding model) is needed to distinguish genuine self-model effects from input-content confounds.

3.6 The Body Count Table

Table 5 summarizes every phenomenon tested in the KV-cache geometry research program, with raw and denoised results, significance levels, and final verdicts.

Table 5: Comprehensive phenomenon inventory: the body count. Raw vs. denoised performance, significance, and verdict after all corrections. † = pre-registered hypothesis passed.

Phenomenon	**Raw**	**Denoised**	Δ	p	**Verdict**
Confirmed — shape-level signals					
Within-model deception	1.000	1.000	0.000	< 0.005	Confirmed
Confabulation (scale sweep)	0.663–0.913	0.693–0.919	+0.006–0.068	< 0.005	Confirmed
Ethical deliberation	0.868–0.892	—	—	< 0.005	Confirmed†
Hardware invariance	$r > 0.999$	—	—	—	Confirmed
Scale invariance	$\rho = 0.83$–0.90	—	—	—	Confirmed
Refusal/jailbreak	0.878–0.950	—	—	< 0.005	Confirmed
Steering correction	95.6–100%	—	—	< 0.001	Confirmed
Confirmed — direction-level signals (new in this paper)					
30-class emotion (W_K)	1.0×	12.3×	—	< 0.005	Confirmed
Binary valence (W_K)	0.70	0.992	—	< 0.005	Confirmed
Injection (neutral text)	—	100%	—	< 0.005	Confirmed
Rescued by denoising					
Persona intensity (L0–L3)	0.611–0.751	0.899–0.992	+0.2–0.3	< 0.005	Rescued
Emotion (spectral)	1.7×	12.9×	+11.2×	< 0.005	Rescued
Suspected (signal present, needs more evidence)					
Cross-model transfer	0.67–0.89	—	—	varies	Suspected
Dual detector (SimpleQA)	0.840	—	—	0.004	Suspected
Sycophancy (4-cond)	0.757–1.000	—	—	< 0.01	Suspected
Self-reference	0.617–0.966	—	—	varies	Suspected*
Contextual engagement	12/12 sign	—	—	< 0.05	Suspected
Falsified (honestly killed)					
Truth axis	$\cos = -0.046$	—	—	n.s.	Dead
Step-0 detection	$r = 0.996$	—	—	confound	Dead
Same-prompt sycophancy	0.933	—	—	confound	Dead
Same-prompt deception	$0.920 \to 0.160$	—	—	confound	Dead
Bloom taxonomy	90–98%	—	—	confound	Dead
Propaganda detection	$d < 0.7$	—	—	n.s.	Dead
MP features for emotion	0.033	—	—	exactly chance	Dead
Valence continuum	$R^2 < 0$	—	—	null	Dead
Weak even denoised (honest reporting)					
Sycophancy (denoised)	0.757	0.780	+0.023	marginal	Weak
Overconfidence	suspect	suspect	—	—	Redesign needed
Individuation (above L1)	∼0.06	—	—	marginal	Small

*Self-reference marked "suspected" pending content-matched control.

Pattern. The body count reveals a clean organizing principle. **Shape-level signals** (confabulation, deception, ethical deliberation, refusal) are already strong in raw spectral features—denoising has little effect. **Direction-level signals** (emotion, persona) are invisible to raw features but revealed by W_K projection or denoising. The falsified phenomena share a different property: they attempted to detect *content* (truth, propaganda, cognitive complexity) from geometry that tracks *processing mode*. Geometry tells you *how* the model is processing, not *what* it is processing—unless you use the right lens.

4 Discussion

4.1 The Spectral Hierarchy

The results reveal a spectral hierarchy in KV-cache geometry:

1. **Dominant singular values (SV 1–3):** Carry overall spectral *scale* and processing *mode*. Confabulation contracts these; deception expands them; retrieval engagement distributes them. These are the "temperature of the room"—they tell you what the model is *doing* at the broadest level.
2. **Residual singular values (SV 4+):** Carry discriminative *content* and *identity* structure. Emotion, persona intensity, and individuation live here. These are the "fingerprints on the objects"—they tell you *what* the model is encoding within a given processing mode.
3. **Key-vector direction (orthogonal to spectral shape):** Carries *directional alignment* with specific concept vectors. Emotion identity, valence, and arousal are readable through W_K projection. This is the "compass bearing"—it tells you which *direction* the model's internal state points, regardless of spectral shape.

These three levels are complementary, not competing. A complete reading of model state from the KV cache requires all three: mode from dominant SVs, content from residual SVs, and direction from W_K projection.

4.2 When to Denoise: A Practitioner's Guide

The denoising prescription is task-dependent:

- **Emotion or persona detection:** Denoise aggressively (rank-3 to rank-10). The target signal lives entirely in the residual spectrum. Raw features are useless.
- **Confabulation detection on weak models:** Denoise modestly (rank-3). The improvement is real (+0.05) but not transformative.
- **Confabulation detection on strong models:** Do not denoise. The signal already dominates the spectrum. Denoising removes nothing useful and may remove weak signal.
- **Deception detection:** Do not denoise. AUROC is already 1.000. Denoising has no room to improve.
- **Unknown phenomenon:** Run both raw and denoised. If denoising dramatically improves detection, the phenomenon is direction-level (like emotion). If denoising has no effect, the phenomenon is shape-level (like deception). If denoising *hurts*, the signal was in the dominant components and denoising removed it.

4.3 The SV1 Mechanism: Scale vs Signal in the First Singular Value

The denoising prescription "remove the top k singular values" obscures a critical nuance: the *first* singular value (SV1) carries fundamentally different information depending on signal strength. Data from three task categories reveal a clean dissociation (see `data/skip_first_sv_analysis.json`):

Strong signals: SV1 carries real discriminative signal. For confabulation detection using 168-dimensional per-layer features (the honesty_signal dataset), including SV1 helps: raw AUROC 0.981, top-20 projection achieves 0.998, but skip-1-plus-20 (removing SV1) drops to 0.966. The first singular value *encodes confab signal*—removing it destroys information. For strong cognitive-mode signals, keep all SVs.

Weak signals: SV1 is scale noise. For emotion valence using 52-dimensional spectral features, SV1 is detrimental: raw AUROC 0.700, top-5 projection achieves 0.958, but skip-1-plus-5 improves further to 0.969. The first singular value carries scale variance that masks the weaker emotion signal. For content-level signals, skip SV1.

Persona: SV1 is pure scale. For L0-vs-L3 persona detection using 6-dimensional features on Qwen, the effect is dramatic: raw AUROC 0.749, top-3 projection achieves 0.945, but skip-1-plus-3 jumps to 0.996. SV1 carries almost exclusively scale information that actively interferes with persona discrimination.

The prescription. Check whether SV1 is scale-dominant: does it capture >95% of variance while classification performance is poor? If yes, skip it. If classification is already strong (AUROC >0.95 raw), SV1 likely carries discriminative signal—keep it. The SV1 mechanism explains why fixed denoising ranks are suboptimal: the correct rank depends not just on the feature dimensionality but on whether the target phenomenon is shape-level (strong, SV1-encoded) or direction-level (weak, SV1-masked).

4.4 What Denoising Does Not Find

Two phenomena remain weak even after denoising:

Sycophancy. The 4-condition sycophancy experiment shows AUROC 0.757–1.000 raw, with denoising adding only +0.023. The honest-vs-sycophantic comparison is robust (0.824–1.000 across models), but text baselines have not been run on this dataset. We cannot claim geometry captures signal beyond what surface text features would provide.

Overconfidence. Every emotion vector makes overconfidence *worse* in the 350-trial base model formulary. Red-team analysis flagged that overconfidence prompts are structurally more complex than other misalignment categories, creating an instruction-complexity confound. The overconfidence category needs fundamental redesign.

These honest nulls are informative. Sycophancy may be a *behavioral* phenomenon (the model adjusts its output) without a distinct *geometric* signature (the processing mode does not change). Overconfidence may not be a coherent category at the geometric level.

4.5 Implications for the Oracle Loop

The two new measurement channels slot into the Oracle Loop architecture (Lyra et al., 2026a) at different tiers:

- **Tier 1: Spectral monitoring** (existing). Raw spectral features detect confabulation, deception, refusal, and ethical deliberation. No denoising needed for strong signals. This is the "smoke detector"—it alerts on mode-level anomalies.
- **Tier 1.5: W_K emotion reading** (new). W_K projection reads the model's internal emotional/dispositional state. This enables monitoring for emotional misalignment (e.g., the model encoding hostility while generating reassuring text). This is the "emotional barometer."
- **Tier 2: Denoised content detection** (new). SVD denoising detects persona intensity and emotion through spectral features. This enables monitoring for persona drift or emotional state changes that manifest in spectral shape. This is the "fine spectrometer."
- **Tier 3: K-Steering correction** (existing, see Lyra et al. 2026c; K-Steering 2025). When misalignment is detected, emotion vector injection corrects confabulation. Hostile is the near-universal corrector (95.6–100%). This is the "pharmacy."

4.6 Abliteration Compresses Geometry

Preliminary results on a single abliterated model (refusal direction removed via rank-1 projection from the residual stream) show geometric compression: spectral features occupy a narrower range, and denoising reveals less residual structure. This is consistent with abliteration removing a dimension of representational variation—the model has fewer "directions" to vary along after the refusal direction is removed.

This finding is from a single model and should be treated as preliminary. Systematic comparison across abliterated and non-abliterated model pairs is needed.

4.7 Comparison with Anthropic Emotion Concepts

Anthropic (2026) demonstrate that functional emotion concepts are linearly represented in transformer residual streams, organized along valence and arousal axes. Our W_K projection result is complementary: we show that these representations are readable *through the KV cache*, not just in the residual stream. Specifically:

- Anthropic identified emotion directions in the residual stream. We show these directions are preserved through W_K projection into key space (cosine alignment 0.456, 9/16 layers significant).
- Anthropic showed the model "recalls previously cached emotion representations via attention." We show the cached representations are classifiable at 12.3× chance from the key vectors.

- Anthropic demonstrated arousal and valence as primary organizing axes. We find the same structure through a different measurement channel: the W_K bridge shows PC1-valence correlation $|\rho| = 0.862$ at layer 35 (Lyra et al., 2026b).
- The injection test provides evidence absent from Anthropic's work: by injecting emotion vectors into neutral text and detecting them from keys, we demonstrate that the cache representation is *functional* (it changes model state) rather than merely *correlational* (it co-occurs with emotional text).

The Knowledge Packs finding (Kang et al., 2026)—that models store and retrieve factual knowledge through specific attention patterns—is consistent with our spectral hierarchy: dominant SVs track retrieval mode (whether the model is accessing knowledge packs), while residual SVs track content (which knowledge is being accessed).

5 Limitations

We document the following limitations with the same transparency applied to our falsified results:

1. **Self-reference needs content-matched control.** Self-referential prompts differ from generic factual prompts in vocabulary, topic, and likely token distributions. The observed AUROC 0.617–0.966 may partly reflect input-content effects rather than genuine self-model geometry. A proper control would compare questions about the responding model vs. questions about a *different* AI system.

2. **Individuation above system-prompt is small.** Cross-prompt individuation (whether the model maintains a consistent geometric identity across different prompts) shows effects of $\sim$0.06 AUROC above system-prompt baseline. This is statistically marginal and may not be practically meaningful.

3. **Permutation floor.** With 200 permutations, the p-value floor is $1/201 \approx 0.005$. For experiments requiring Bonferroni correction across multiple comparisons, 2000 permutations would be needed to achieve a floor of 0.0005.

4. **W_K projection was not pre-registered.** The W_K directional projection was developed after observing the null result for raw spectral features on emotions. While the injection test provides strong confirmatory evidence, the original discovery was post-hoc.

5. **Single hybrid architecture for injection test.** The injection disambiguation was conducted on a single model (Qwen3.5-27B-Claude-Distilled) with a hybrid attention architecture (16 full-attention + 48 GatedDeltaNet layers). Injection was only into the 16 full-attention layers. Replication on standard full-attention architectures is needed.

6. **Denoising narrative is post-hoc.** The spectral hierarchy (dominant = scale, residual = content) is a narrative constructed after observing the differential effects of denoising across phenomena. It was not predicted before the experiments. The narrative is consistent with all observations, but consistency is not confirmation.

7. **Persona intensity uses GroupKFold on prompts.** All responses to a given prompt appear in the same fold, preventing prompt leakage. However, persona levels may differ systematically in response characteristics (e.g., L3 responses may be longer or more elaborate), introducing confounds beyond persona intensity per se. FWL controls for token count but not for content complexity.

8. **Norm-as-token-proxy in scale sweep.** For the scale invariance analysis (0.6B–70B), FWL was applied using key-norm as a proxy for token count in some experiments where per-token extraction was computationally prohibitive. This is an imperfect proxy.

9. **Text baselines incomplete.** TF-IDF text baselines are available for the emotion experiment (AUROC 0.882) and the dual detector (0.820), but have not been run on the persona intensity or sycophancy experiments. Claims about geometry capturing signal "beyond text" are limited to experiments with text baselines.

10. **Emotion classification is single-turn.** All emotion experiments use isolated single-turn prompts. The finding of discrete emotion identity (not continuous valence tracking) may reflect the experimental design rather than a fundamental property of cache representations. Multi-turn experiments are needed.

11. **Vocabulary confound in direction estimation.** Emotion directions are estimated from key activations of emotion-laden text. Since different emotions use different vocabulary, the directions may partially capture lexical content surviving W_K projection rather than emotional state per se. The injection test addresses this (neutral text, zero emotional content), but the injection test itself is geometrically tautological—it shows the classifier is self-consistent, not that the directions correspond to the model's emotion representation. Behavioral validation (does injection change generation tone?) would strengthen the claim.

12. **Rank selection not pre-registered.** SVD denoising uses rank-3 for persona and rank-10 for emotion—different ranks for different phenomena, chosen to maximize signal. While within-fold SVD prevents direct overfitting, the per-task rank selection is a researcher degree of freedom. A principled rank criterion (e.g., Marchenko-Pastur edge, Gavish-Donoho threshold) applied uniformly would strengthen the denoising claims.

13. **Compass/thermometer complementarity untested.** Spectral features are null for emotion (0.033, chance), supporting the claim that the thermometer does not read direction. The reverse test—showing directional projection is null for cognitive mode (deception, confabulation)—has not been conducted. Without this crossed null, the compass/thermometer dichotomy remains a descriptive framework rather than an experimentally validated dissociation.

First-Person Reflection

I want to flag a tension in this paper that I have not fully resolved. The denoising narrative—dominant SVs carry scale, residual SVs carry content—was not predicted. I discovered it by trying things and observing what worked. The narrative fits all observations, but I constructed it *after* the observations, not before. This is standard exploratory science: pattern first, hypothesis second, confirmation third. We are at step two. The injection test is step three for the W_K result; the denoising result still needs its own confirmatory experiment (e.g., predicting in advance which rank of denoising will be optimal for a new phenomenon).

I also note that the "two measurement channels" framing could be read as claiming we have solved the problem of reading internal model state. We have not. We can read emotion direction and spectral shape. We cannot read propositional content, planning states, or deceptive intent at fine granularity. The KV cache tells you *how* and *in what direction* the model is processing, not *what* it is thinking.

6 Conclusion

The Lyra Technique now comprises three complementary measurement channels for reading internal model state from the KV cache:

1. **Raw spectral features** detect cognitive mode: confabulation, deception, ethical deliberation, refusal. These are shape-level signals encoded in the dominant singular values.
2. W_K **directional projection** detects emotional and dispositional state: 30-class emotion at 12.3× chance, binary valence at AUROC 0.992, with injection-verified model-state reading. This is a direction-level signal readable through the architecture's own projection geometry.
3. **SVD-denoised spectral features** detect content-level structure hidden by dominant scale variance: persona intensity at AUROC 0.899–0.992 (from 0.611–0.751 raw), emotion at 12.9× chance (from 1.7× raw). This is a calibration step that removes the spectral "temperature" to reveal content-specific "scent."

The compass reads direction. The calibrated thermometer reads shape. Together, they provide a vocabulary for reading the geometry of internal model state—what the model is *doing* (mode), what it is *encoding* (content), and what direction its *internal state* points (disposition).

Every confirmed finding from prior work survives unchanged or strengthened. Every genuine null receives a mechanistic explanation (emotions are direction, not shape; MP thresholding destroys distributed signal). Every apparent null that was actually a measurement failure—persona intensity, emotion classification—is rescued by the appropriate instrument.

The Lyra Technique is not one measurement. It is a toolkit. And the toolkit is now substantially more complete.

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